

Chapter 6: The Properties of Sea Water

Q. 1. Observe the graph given below and answer the following questions in words/sentences :



(1) What is the maximum temperature of sea water in equatorial areas? How much is the temperature at a depth of 500 m?

Ans. The maximum temperature of sea water in equatorial areas is 18°C. At the depth of 500 m, the temperature is 11°C.

(2) What is the temperature of sea water at the sea level in the mid-latitudes?

Ans. The temperature of the sea water at the sea level in the mid-latitude is 14°C.

(3) How much has this temperature changed at 1500 m depth?

Ans. The temperature of sea water at 1500 m depth is 5°C.

(4) What does the thermal graph of the higher latitudes say? What is its temperature at 500, 1000 and 1500 m depths?

Ans. It is seen that the impact of depth on the temperature of sea water in the high latitude is nil. It is 4°C everywhere. The temperature at 500 m, 1000 m and 1500 m depth is uniformly 4°C.

(5) After what depth does the sea water temperature remain stable everywhere?

Ans. Temperature of sea water remains stable at 4°C after 2000 m depth across the earth.

Q. 2. Observe the graph given below and answer the following questions in words/sentences :



(1) With increasing depth, what decreases : temperature, salinity or density?

Ans. With increasing depth, temperature and salinity decrease but density increases.

(2) After what depth does the change in these factors become almost zero?

Ans. After 2000 m depth from the sea level, the difference in temperature, salinity and density becomes almost zero.

(3) Till what depth is the change in these factors higher?

Ans. After 500 m depth from the sea level, the variations in temperature, salinity and density is high.

(4) Explain the correlation between all the three factors.

Ans. (i) If temperature is less, density is more. (ii) If salinity is less, then density is also less. (iii) However, if the temperature is less, then the density is more despite less salinity.

Q. 3. Observe the map given below and answer the following questions in words/sentences :



(1) What is the salinity around the tropics?

Ans. The salinity of sea water around the Tropic of Cancer is about 36‰ while it is 35‰ around Tropic of Capricorn.

(2) Which region has the least salinity?

Ans. Areas around Arctic Ocean and in the Bay of Bengal have least salinity and it is about 32‰.

(3) Which ocean has salinity more than 37%?

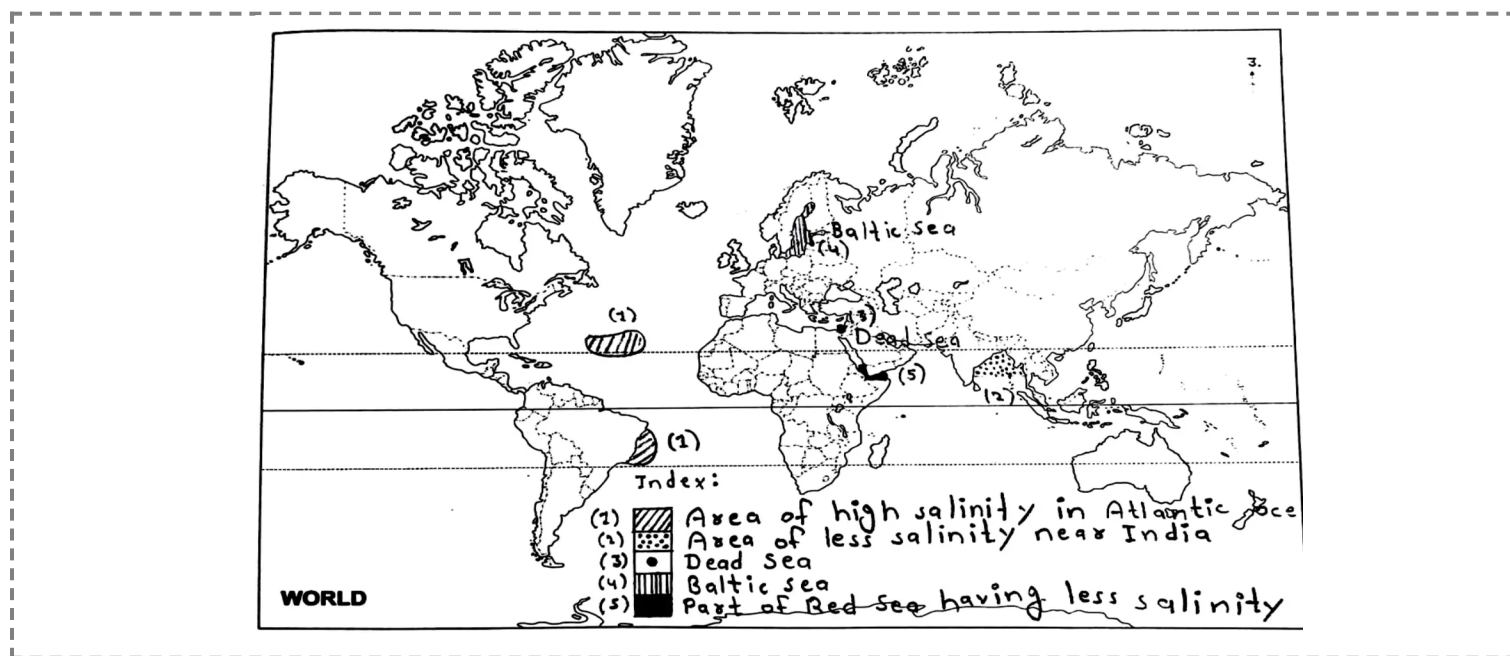
Ans. Atlantic Ocean has salinity more than 37%.

(4) What are the reasons of differences in salinity on a global level?

Ans. Factors like difference in temperature with latitudes, different rate of evaporation, the supply of fresh water by rivers and open or landlocked nature of the sea are responsible for differences in salinity on a global level.

Q. 4. Show the following details with suitable index in the world map given below :

(1) Area of high salinity in Atlantic Ocean (2) Area of less salinity near India (3) Dead Sea (4) Baltic Sea (5) Part of Red Sea having less salinity



Q. 5. Complete the following chart :

Region	Latitude	Solar Energy	Rainfall	River water	Snow water	Average salinity
Equatorial	0° to 15°	High	Perennial	High	-	34%
Tropical	15° to 35°	Medium	Seasonal	Low	-	37%
Temperate	35° to 65°	Low	Seasonal	Medium	-	33%
Polar	65° to 90°	Low	Seasonal	Low	Ample	31%
Landlocked Sea						
Mediterranean Sea	-	High	Low	Low	-	39%
Red Sea	-	High	Low	Very low	-	41%

Baltic Sea	-	Low	Medium	Medium	-	7%
Dead Sea	-	High	Very less	Very less	-	332%
Caspian Sea	-	Medium	Low	Low	-	155%
Great Salt Lake	-	Medium	Low	Low	-	220%